

Prompt Engineering Challenge

1. Summarising Notes

Prompt:

Summarise the following notes into simple bullet points for revision:

Artificial Intelligence (AI) enables machines to perform tasks that usually require human intelligence. Machine Learning is a branch of AI that learns from data. Generative AI can create new content such as text, images, and audio.

AI Response:

- *Artificial Intelligence helps machines perform intelligent tasks.*
- *Machine Learning is a part of AI that learns from data.*
- *Generative AI can create text, images, and audio.*
- *AI is used in many real-world applications.*

Why the Prompt Worked:

The prompt provided the source notes and clearly requested a bullet-point summary, making it easy to generate a response.

2. Generating a Study Plan

Prompt:

Create a 7-day study plan for learning Python basics for a beginner student.

AI Response:

Day 1: *Introduction to Python and installation*

Day 2: *Variables, data types, and input/output*

Day 3: *Conditional statements (if, else)*

Day 4: Loops (for and while)

Day 5: Functions and basic problem solving

Day 6: Lists, dictionaries, and practice exercises

Day 7: Revision and a small Python project

Why the Prompt Worked:

The prompt clearly mentioned the subject (Python), the duration (7 days), and the target audience (beginner student). This helped generate a structured and relevant study plan.

3.Creating Interview Questions

Prompt:

Generate 5 beginner-level Python interview questions with clear and professional answers.

AI Response:

1.What is Python?

Answer:

Python is a high-level, interpreted programming language known for its simple syntax and readability. It is widely used in web development, data science, artificial intelligence, and automation.

2.What is a variable in Python?

Answer:

A variable is a named storage location used to hold data values. Variables allow programs to store and manipulate information during execution.

3.What is the difference between a list and a tuple?

Answer:

A list is mutable, meaning its contents can be changed after creation. A tuple is immutable, meaning its contents cannot be modified once created.

4.What is a function in Python?

Answer:

A function is a reusable block of code designed to perform a specific task. Functions help improve code organization and reduce repetition.

5.What is a loop in Python?

Answer:

A loop is a programming structure used to execute a block of code repeatedly until a specified condition is met or for a defined number of iterations.

Why the Prompt Worked: *The prompt clearly specified the subject (Python), the experience level (beginner), and the requirement for professional answers. This resulted in clear, structured, and relevant interview questions.*

4.Writing a Resume Summary

Prompt:

Write a short resume summary for a computer science student interested in AI and software development.

AI Response:

Computer Science student with a strong interest in Artificial Intelligence and Software Development. Familiar with Python programming, problem-solving, and basic software development concepts. Eager to learn new technologies and apply technical skills to real-world projects.

Why the Prompt Worked:

The prompt clearly described the student's background and career interests. This helped generate a concise and relevant resume summary suitable for a beginner-level profile.

5.Creating a Lesson Plan

Prompt:

Create a one-hour lesson plan for revising basic Java programs for an interview.

AI Response:

Duration: 1 Hour

0–10 Minutes:

- *Introduction and interview tips*
- *Review Java syntax and program structure*

10–20 Minutes:

- *Program to check whether a number is even or odd*
- *Program to find the largest of three numbers*

20–30 Minutes:

- *Program to reverse a number*
- *Program to check whether a number is a palindrome*

30–40 Minutes:

- *Program to calculate factorial using a loop*
- *Program to generate the Fibonacci series*

40–50 Minutes:

- *Program to find prime numbers*

- *Program to sort an array in ascending order*

50–60 Minutes:

- *Quick coding practice and discussion of common interview questions*

Why the Prompt Worked:

The prompt clearly specified the subject (Java), the purpose (interview revision), and the duration (one hour). This helped generate a focused lesson plan containing commonly asked programming problems.